

Voice Agent System

Telephony Price Comparison & Technical Report

Forage AI · June 17, 2026 · Prepared by Salomi Mekala

Doctor Branch Discovery · Automated Outbound Calling · Indian Hospitals

1. What We Built

Voice Agent — Automated Hospital Branch Finder

The Voice Agent automatically calls Indian hospitals, holds a natural spoken conversation with the receptionist, and extracts which **branch or city** a specific doctor is based at — then saves the result to the database. No human involvement required.

System Context

This is part of an 8-agent pipeline. Forage AI's team owns **Agent 3 (Email)** and **Agent 6 (Voice)**. The Voice Agent dials hospitals and extracts doctor branch information.

Hardware Constraints

Component	Spec	Impact
CPU	Intel i5-1155G7	All AI models run on CPU — no GPU acceleration
RAM	16 GB	Limits model size — must use small models
OS	Windows 11	No Docker — all native installs
GPU	Intel Iris Xe (integrated)	No CUDA — cannot use GPU for AI
Requirement	Open-source only	No paid AI APIs — Whisper, Piper, Ollama

2. How a Call Works

Call Flow — Step by Step

Step	Component	What Happens
1	run_twilio.py starts	Warms up Whisper + Piper models, starts FastAPI server on port 8000
2	Twilio REST API	Script tells Twilio: call this hospital number (+91), webhook = ngrok URL
3	ngrok tunnel	Twilio sends audio/events to public ngrok URL → forwarded to localhost:8000
4	Hospital picks up	Twilio plays trial warning (paid = no warning), hospital presses key
5	POST /answer	Twilio hits our server → we return TwiML: open WebSocket stream
6	WebSocket opens	Live two-way audio pipe between Twilio and our server
7	Conversation loop	Hospital speaks → Whisper → Brain → Piper → plays back to hospital
8	Branch saved	save_branch() stores result, call ends, recording saved to disk

Role of ngrok

Your laptop sits behind a home WiFi router with a private IP (e.g. 192.168.1.5) that the internet cannot reach. **ngrok** creates a secure encrypted tunnel giving your laptop a public URL (e.g.

https://protozoan-undoing-rug.ngrok-free.dev). All Twilio webhook requests go through this URL → tunnel → your server. If ngrok is not running, Twilio cannot reach your server and plays 'Application error, goodbye'.

Command: ngrok http 8000 (run in a separate terminal before the script)

3. Technical Components

Component	Technology	Role	CPU Time
Speech-to-Text	faster-whisper (tiny + small)	Converts hospital speech to text	~350ms
Brain / AI	Heuristic + phi4-mini via Ollama	Decides what to say next	2ms–10s
Text-to-Speech	Piper TTS ONNX en_US-lessac-high	Converts agent reply to audio	~350ms (0ms cached)
Phone calls	Twilio Media Streams + WebSocket	Dials hospital, streams audio	N/A
Tunnel	ngrok	Exposes localhost to internet for Twilio	~150ms network
Server	FastAPI + uvicorn	Handles webhooks and WebSocket audio	<5ms
Recording	soundfile + numpy	Saves call audio + transcript + JSON	<1s after call

Latency Breakdown per Turn

Stage	Time	Can Reduce?	How
Silence detection	500ms	Yes	Reduce frames (risk: cuts off speech early)
Whisper tiny STT	350ms	Partial	GPU would make it 20ms
Brain (heuristic)	2ms	N/A	Already instant — pure Python
Brain (LLM)	5–10s	Yes	Expand heuristics to avoid LLM calls
Piper TTS	350ms	Yes	Pre-cache = 0ms for fixed phrases
Network (Twilio)	150ms	No	Fixed internet latency
TOTAL (heuristic)	~1.0s	Yes	~0.65s with full cache on GPU

Note: Twilio trial vs paid has ZERO effect on latency. All lag comes from CPU processing. A cloud GPU (e.g. Runpod RTX 3080 at \$0.20/hr) would reduce total latency to ~300ms.

4. Telephony Providers Evaluated

US / Global Providers — No Monthly Plan, Pay-As-You-Go

Provider	Number/mo	Outbound India/min	Free Trial	KYC?	WebSocket ?	Status
Twilio ■■	\$1.15	\$0.022	\$14.35	No	Yes	Current ■
Telnyx ■■	\$1.00	\$0.016	\$10	No	Yes	Cheapest US
SignalWire ■■	\$1.15	\$0.018	\$5	No	Yes	In .env
Plivo ■■	\$0.80	\$0.018	\$2	No	Yes	Login issues
Vonage ■■	\$1.40	\$0.025	€2	No	Yes	Not tried

All US providers are pure pay-as-you-go — no monthly plan, no commitment. Just add credit and pay per minute used. Best for testing and low volume.

Indian Providers — Monthly & Annual Plans (4× Cheaper per Minute)

Provider	Starter/mo	Annual Plan	Annual Saving	India/min	KYC?	Status
Exotel ■■	■2,499 (~\$30)	■24,990/yr	Save 17%	■0.50	Yes	KYC failed
Ozonetel ■■	■3,000 (~\$36)	■30,600/yr	Save 15%	■0.50	Yes	Not tried
Servetel ■■	■1,999 (~\$24)	■19,990/yr	Save 17%	■0.45	Yes	No free
MyOperator ■■	■3,000 (~\$36)	■28,800/yr	Save 20%	■0.55	Yes	Not tried
Knowlarity ■■	■3,500 (~\$42)	■33,600/yr	Save 20%	■0.50	Yes	Not tried
Airtel IQ ■■	■5,000 (~\$60)	■48,000/yr	Save 20%	■0.40	Yes	Enterprise

Indian providers require TRAI compliance (KYC): company GST, PAN, business registration, DLT registration. Approval takes 3–7 days. Cannot skip — Indian government regulation.

5. Exotel Plan Details (Recommended Indian Provider)

Plan	Monthly	Annual	Included Mins/mo	Overage Rate	Numbers	Best For
Starter	■2,499 (~\$30)	■24,990 (~\$300)	500	■0.50	1	Testing <500 calls/mo
Growth	■4,999 (~\$60)	■47,990 (~\$576)	1,500	■0.48	2	1,000–3,000 calls/mo
Business	■9,999 (~\$120)	■95,990 (~\$1,152)	4,000	■0.45	5	3,000–8,000 calls/mo
Enterprise	Custom	Negotiated	10,000+	■0.35	∞	8,000+ calls/mo

Annual plans save 15–20% over monthly. Example: Exotel Growth monthly = ■4,999 × 12 = ■59,988. Annual = ■47,990. Saving = ■11,998 (~\$144/year) just by paying upfront.

6. Monthly Cost Comparison — 1,000 Calls/Month (2 min avg = 2,000 min)

Provider	Type	Number Fee/mo	2,000 min Cost	Total/mo	Total/yr	vs Twilio
Twilio	US Pay/use	\$1.15	\$44.00	\$45.15	\$542	—
Telnyx	US Pay/use	\$1.00	\$32.00	\$33.00	\$396	-\$146/yr
SignalWire	US Pay/use	\$1.15	\$36.00	\$37.15	\$446	-\$96/yr
Plivo	US Pay/use	\$0.80	\$36.00	\$36.80	\$442	-\$100/yr
Exotel	India Monthly	Included	■500 extra	~\$36	~\$432	-\$110/yr
Exotel	India Annual	Included	■500 extra	~\$30	~\$360	-\$182/yr

* Exotel Growth plan includes 1,500 min/mo free. Extra 500 min = 500 × ■0.48 = ■240 (~\$3). Prices indicative.

7. US vs Indian Providers — Key Differences

Factor	■ ■ US Providers	■ ■ Indian Providers
Setup time	5 minutes — card only	3–7 days — KYC + approval
KYC required	No	Yes — GST, PAN, business docs
Call rate India	\$0.016–0.025/min	\$0.004–0.007/min (4× cheaper)
Monthly minimum	None — pay as you go	■ 2,000–5,000/mo base plan
Caller ID shown	+1 US number (foreign)	+91 Indian number (local)
Hospital pickup	Lower — looks foreign	Higher — local number trusted
Call quality	International routing	Local routing — better quality
Annual plans	Not available	Yes — 15–20% discount
WebSocket/Stream	Full support	Exotel, Airtel IQ support it
Free trial	Yes — \$2 to \$14 credit	None
Best for	Testing / quick start	Production at scale

8. Recommendation

Stage	Action	Cost	Why
Right Now	Upgrade Twilio trial → paid (add \$50 credit)	\$50 one-time	Removes 'press any key' message. 5 minutes to set up.
Short Term	Try Telnyx (\$10 free credit) — same code, just .env change	\$10 trial	Cheapest US rate (\$0.016/min). No KYC. Easy switch.
Production	Exotel Growth Annual — after Forage AI KYC complete	■ 47,990/yr (~\$576)	Indian number = better pickup rate. 4× cheaper per minute.
GPU Scale	Runpod RTX 3080 during calling hours (9am–6pm IST)	~\$1.80/day (~\$54/mo)	Drops latency from ~1s to ~300ms. True production grade.

Switching from Twilio to any other provider = changing 3 lines in .env file. No code changes required.

9. Bugs Fixed in Day 2

Bug	Fix
Dr. Dr. John in greeting	_clean_name() strips existing Dr. prefix in prompts.py

Echo — agent heard itself	speaking flag blocks incoming audio while TTS plays
Cardiology saved as branch	Specialization blacklist in save_branch()
Oh → agent says goodbye	Check save_branch return value before ending call
Nothing over saved as branch	_NEGATIVE_WORDS check — any negative word = no branch
10-second cold-start latency	Pre-warm Whisper + Piper before placing call
23-second LLM wait	Expanded heuristics + 5s LLM timeout + fallback re-ask
Are you talking about Dr. John?	Added to _NAME_CONFIRMATION_PHRASES — heuristic catches it
Dallas transcribed as Ballard	Added Indian city names as initial_prompt to Whisper
Which branch? from caller	New _CLARIFICATION_PHRASES heuristic response
TTS delay on every turn	Pre-cache all fixed responses at session start — 0ms
Silence cuts too early	_SILENCE_FRAMES reduced 50→25 (1000ms→500ms)

All code is in C:\Users\salom\OneDrive\Desktop\conversational_ai\ · Run with: python run_twilio.py --doctor 'Name' --hospital 'Apollo' --to '+91XXXXXXXXXX'